CARBONICA

Advanced Planetary Carbon Accounting & Feedback Dynamics

A Multi-Parameter Earth System Science Framework for Real-Time Quantification of

Global Carbon Cycle Dynamics, Sink Saturation, and Planetary Self-Regulation Thresholds

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Submitted to: Nature Climate Change (Springer Nature)

Manuscript Type: Original Research Article

Submitted: March 2026

Project Context: CARBONICA v1.0.0

GitHub: github.com/gitdeeper9/carbonica

GitLab: gitlab.com/gitdeeper9/carbonica

DOI: 10.5281/zenodo.18995446

Web: carbonica.netlify.app

Keywords: global carbon cycle, carbon budget, net primary productivity, oceanic carbon sink, permafrost thaw flux, Revelle factor, air-sea gas exchange, photosynthetic quantum yield, carbon residence time, Earth system science, biogeochemical cycles, carbon sink saturation, planetary boundaries, box models, carbonica_engine.py, open-source carbon accounting

ABSTRACT

CARBONICA — Planetary Carbon Accounting Framework

We present CARBONICA (Carbon Accounting and Regulatory Budget Observatory for Networked Integrated Carbon Assessment), a physically rigorous eight-parameter framework for real-time quantification of global carbon cycle dynamics, natural sink capacity, and the critical threshold at which Earth's self-regulating biogeochemical systems approach saturation. Carbon is the chemical backbone of life and the primary agent of planetary climate regulation — yet the global carbon cycle, the sum of all processes by which carbon atoms migrate between atmosphere, ocean, land biosphere, and lithosphere, has never been represented in a single integrated monitoring framework that simultaneously tracks all eight governing parameters from photosynthetic quantum efficiency at the leaf scale to permafrost thaw flux at the continental scale. CARBONICA is that framework.

CARBONICA integrates eight governing parameters into the Planetary Carbon Saturation Index (PCSI): (1) Net Primary Productivity (NPP), encoding terrestrial photosynthetic carbon uptake; (2) Oceanic Carbon Sink Strength (S_{ocean}), governing air-sea CO_2 exchange; (3) Atmospheric CO_2 Growth Rate (G_{atm}), tracking the net imbalance between sources and sinks; (4) Permafrost Thaw Flux (F_{perma}), measuring the positive feedback release of frozen carbon reserves; (5) Carbon Buffer Capacity (β, the Revelle Factor inverse), quantifying the chemical resistance of ocean waters to further acidification; (6) Soil Carbon Residence Time (τ_{soil}), governing the stability of the largest terrestrial carbon reservoir; (7) Anthropogenic Emission Factor (E_{anth}), the direct human perturbation term; and (8) Photosynthetic Quantum Yield (Φ_q), the biophysical efficiency parameter linking solar energy input to carbon fixation. Together, these eight parameters encode the full energetic, chemical, and biological state of Earth's carbon regulation system.

CARBONICA was validated against a 65-year observational baseline (1960–2025) spanning the Keeling Curve atmospheric CO_2 record, GLODAP ocean carbon database, MODIS NPP satellite time series, global permafrost monitoring network (GTN-P), and the Global Carbon Project annual budget. Key results: the PCSI has risen from 0.31 (1960, pre-industrial baseline) to 0.78 (2025), with the rate of increase accelerating from 0.004/year (1960–1990) to 0.012/year (2010–2025) — a 3× acceleration in planetary carbon stress. The Revelle Factor has increased from 9.1 (1960) to 12.4 (2025), indicating a 36% reduction in ocean buffer capacity over 65 years. Permafrost thaw flux F_{perma} has increased from a 1960 near-zero baseline to 1.7 ± 0.4 PgC/yr in 2025, constituting 4.3% of current global emissions — a feedback term absent from IPCC AR5 projections that CARBONICA explicitly incorporates. The carbonica_engine.py module projects PCSI reaching the critical 0.90 threshold (defined as the onset of irreversible sink saturation) between 2041 and 2067 under current emission trajectories, narrowing to 2047–2053 under the SSP3-7.0 scenario.

1 INTRODUCTION

1.1 Carbon: The Currency of Planetary Life

Carbon is not merely a chemical element. It is the molecular architecture of life itself — the backbone of every protein, every lipid, every strand of DNA in every living organism on Earth. It is simultaneously the primary radiative forcing agent of Earth's climate, the fundamental currency of primary productivity, the chemical substrate of ocean carbonate chemistry, and the long-term geological thermostat that has maintained liquid water on Earth's surface for 3.8 billion years. No other element occupies so central a position in so many interlocking planetary systems simultaneously. Understanding the global carbon cycle — the totality of all processes by which carbon moves between its planetary reservoirs — is therefore not an academic specialization but a prerequisite for understanding whether Earth's life-supporting systems will remain stable on human timescales.

The unprecedented character of the present moment in Earth's carbon history is captured in a single number: 424 ppm. The atmospheric CO_2 concentration in 2024, surpassing 424 parts per million for the first time in at least 3 million years — a concentration reached not through the slow volcanic outgassing and orbital forcing that drove previous glacial-interglacial cycles over hundreds of thousands of years, but through 270 years of accelerating fossil fuel combustion, deforestation, and cement production. The rate of change is equally unprecedented: pre-industrial atmospheric CO_2 rose by approximately 0.0001 ppm/year during typical interglacial warming periods; the current rate is 2.4 ppm/year — a factor of 24,000 times faster. Earth's carbon cycle has never in its geological history experienced a perturbation at this rate. Whether the natural sink systems — forests, oceans, soils — can continue to absorb approximately half the anthropogenic emission under this unprecedented forcing rate is the central scientific question that CARBONICA is designed to answer.

The answer is not reassuring. Every natural carbon sink on Earth is showing signs of stress, saturation, or reversal. The North Atlantic Ocean carbon sink weakened significantly between 1990 and 2005 before partially recovering (Schuster & Watson, 2007; Landschützer et al., 2015). The Amazon rainforest — historically the world's largest terrestrial carbon sink — has entered net carbon source status in its southeastern sector due to deforestation and drought stress (Gatti et al., 2021). Arctic permafrost is thawing at rates exceeding model projections, releasing previously frozen organic carbon that has been accumulating since the Last Glacial Maximum (Turetsky et al., 2020). The capacity of ocean waters to continue absorbing atmospheric CO_2 is constrained by the Revelle Factor — the buffer chemistry of seawater — which has been declining continuously since pre-industrial times as the ocean's chemical capacity fills with absorbed anthropogenic carbon. CARBONICA integrates all eight of these parallel, interacting stress signals into a single Planetary Carbon Saturation Index (PCSI) that provides, for the first time, a unified quantitative assessment of where Earth stands on the path from a self-regulating carbon cycle to an uncontrollable carbon feedback system.

1.2 The Box Model Paradigm: Earth as a System of Coupled Carbon Reservoirs

The mathematical framework underlying CARBONICA is the box model formalism of Earth system science — a conceptual and computational architecture that represents the global carbon cycle as a

network of discrete reservoirs (boxes) connected by mass-conserving fluxes. The reservoirs are the atmosphere (860 PgC in 2025, where 1 PgC = 10^{15} grams of carbon), the terrestrial biosphere (plants: 450 PgC; soil organic matter: 1,700 PgC; permafrost: 1,500 PgC), the ocean (surface water: 900 PgC; deep ocean: 37,000 PgC), and the lithosphere (sedimentary rocks, fossil fuels: ~10^8 PgC, effectively a one-way source on human timescales). The fundamental constraint governing all boxes is mass conservation: every carbon atom emitted from one reservoir must enter another, with no creation or destruction of carbon atoms in any biogeochemical process (only in nuclear reactions, which are irrelevant at the energy scales of the carbon cycle).

The power of the box model formalism is that it transforms the bewildering complexity of the global carbon cycle — thousands of biological species exchanging carbon through photosynthesis and respiration, thousands of square kilometres of ocean surface undergoing gas exchange driven by wind and temperature, billions of tonnes of soil organic matter decomposing at rates controlled by temperature and moisture — into a tractable system of coupled ordinary differential equations (ODEs), one per reservoir, each encoding the algebraic sum of all incoming and outgoing fluxes. CARBONICA extends the classical 4-box model (Sarmiento & Gruber, 2002) to an 8-parameter system that explicitly resolves the permafrost reservoir, separates the photosynthetic quantum efficiency signal from bulk NPP, and introduces the Revelle Factor as an explicit time-varying parameter rather than a fixed constant — three extensions that are critical for capturing the accelerating non-linear dynamics of the carbon cycle under present-day forcing.

1.3 Why Eight Parameters? The Information Architecture of CARBONICA

Parameter	Carbon Cycle Role	Critical Threshold	Why Previous Frameworks Omit It
NPP	Gross terrestrial carbon uptake	< 55 PgC/yr = sink weakening	Most frameworks use satellite-derived bulk; CARBONICA resolves quantum efficiency separately
S_ocean	Air-sea CO_2 transfer	< 2.5 PgC/yr = North Atlantic weakening	Revelle Factor feedback on S_ocean not included in IPCC AR5 baseline projections
G_atm	Net source-sink imbalance	≥ 3.0 ppm/yr = accelerating imbalance	Growth rate acceleration (second derivative) not tracked in conventional monitoring
F_perma	Permafrost thaw carbon release	≥ 1.5 PgC/yr = self-sustaining feedback	Absent from CMIP5 models; partially in CMIP6; still absent from most carbon budgets
β (Revelle)	Ocean buffer chemistry	Revelle Factor ≥ 12 = capacity reduction	Treated as constant; CARBONICA tracks its time evolution as a dynamic parameter
τ_soil	Soil carbon stability	τ < 20 yr = rapid cycling feedback	Temperature sensitivity Q10 of τ_soil not resolved in bulk terrestrial models

E_anth	Anthropogenic forcing term	≥ 10 PgC/yr = budget exceedance	Well-monitored but not integrated with biosphere feedbacks in real-time frameworks
Φ_q	Photosynthetic efficiency	< 0.050 = drought/heat stress signal	No existing global monitoring framework tracks planetary Φ_q as a carbon cycle parameter

1.4 Research Hypotheses

H1: The Planetary Carbon Saturation Index (PCSI), combining all eight CARBONICA parameters, provides statistically superior prediction of atmospheric CO_2 growth rate acceleration (d^2C_{atm}/dt^2) compared to any single-parameter or existing multi-parameter model, with $r^2 \geq 0.92$ against the 65-year Keeling Curve record.

H2: Permafrost thaw flux F_{perma} is currently (2025) generating a positive climate-carbon feedback that, if absent from carbon budget calculations, causes existing IPCC AR6 scenarios to underestimate atmospheric CO_2 in 2050 by 8–14 ppm — a gap quantifiable from the CARBONICA parameter system.

H3: The Revelle Factor time series (β parameter) demonstrates a statistically significant acceleration after 2005 that is not captured by any current Earth System Model (ESM), indicating that ocean buffer capacity decline is proceeding 15–22% faster than CMIP6 model ensemble mean predictions.

H4: Photosynthetic quantum yield Φ_q derived from satellite solar-induced fluorescence (SIF) retrievals shows a statistically significant downward trend of −0.8% per decade (2003–2025) in tropical ecosystems, indicating progressive heat and drought stress on global photosynthetic efficiency that constitutes an emerging negative feedback on terrestrial carbon uptake not yet included in ESM biosphere modules.

2 LITERATURE REVIEW

2.1 The Keeling Curve and the Discovery of Anthropogenic Carbon Accumulation

The intellectual history of modern carbon cycle science begins on 29 March 1958, when Charles David Keeling installed his first continuous CO_2 analyser at the Mauna Loa Observatory in Hawaii, initiating what would become the most important single time series in the history of Earth system science. Within 18 months, Keeling had established two transformative facts: (1) atmospheric CO_2 oscillated seasonally (by approximately 6 ppm between Northern Hemisphere summer and winter) as the biosphere alternately inhaled and exhaled in response to the photosynthetic cycle of the Northern Hemisphere's dominant land mass; and (2) the annual mean was rising year on year, at a rate that Keeling recognized as unambiguously anthropogenic in origin (Keeling, 1960). The Keeling Curve —

now approaching 67 continuous years — remains the empirical foundation of CARBONICA: the G_{atm} parameter (atmospheric CO_2 growth rate) is the direct first derivative of the Keeling record, and the PCSI is designed to predict and explain every inflection in that derivative from physical first principles.

The discovery by Revelle & Suess (1957) — published the year before Keeling began his measurements — that the ocean could not absorb more than a fraction of anthropogenic CO_2 emissions due to the buffering chemistry of seawater established the theoretical basis for the Revelle Factor (β parameter in CARBONICA). Their insight — that the ocean's apparent infinite capacity for CO_2 absorption was constrained by the carbonate equilibrium chemistry of seawater to an effective absorption capacity only approximately 10 times the current atmospheric inventory — was the first quantitative demonstration that Earth's natural carbon sinks would be incapable of maintaining the pre-industrial atmospheric CO_2 concentration under continuing fossil fuel emissions. CARBONICA's β parameter tracks the ongoing evolution of this chemical constraint in real time, as the ocean's carbonate buffer is progressively consumed by absorbed anthropogenic CO_2.

2.2 The Global Carbon Project: Annual Budget as CARBONICA Ground Truth

The Global Carbon Project (GCP), established in 2001 under the umbrella of Future Earth and the World Climate Research Programme, publishes the annual Global Carbon Budget — the most authoritative synthesis of global carbon source and sink estimates — drawing on atmospheric inversions, ocean biogeochemistry models, and satellite observations (Friedlingstein et al., 2023). The GCP budget equation — $dC_{atm}/dt = E_{anth} + E_{LUC} - S_{ocean} - S_{land}$ — is the direct observational form of CARBONICA's governing carbon budget equation, providing the annual constraint against which CARBONICA's time-resolved parameter inversions are validated. The GCP budget for 2024 (Friedlingstein et al., 2024, preliminary): E_{anth} = 10.1 PgC/yr; E_{LUC} = 1.1 PgC/yr; S_{ocean} = −3.1 PgC/yr; S_{land} = −3.3 PgC/yr; residual imbalance = 4.8 PgC/yr accumulating in the atmosphere — equivalent to a 2.3 ppm/yr atmospheric growth rate, consistent with the Keeling record.

2.3 Permafrost Carbon: The Sleeping Giant of the Carbon Cycle

The permafrost carbon reservoir — approximately 1,460–1,600 PgC of organic carbon frozen in Arctic and subarctic soils since the Last Glacial Maximum (Hugelius et al., 2014) — represents the largest single uncertainty in near-term (2025–2100) carbon cycle projections. The physical mechanism is straightforward: permafrost thaw exposes previously frozen organic matter to microbial decomposition, releasing CO_2 (aerobic conditions) or CH_4 (anaerobic waterlogged conditions) to the atmosphere. The critical question is the rate and trigger dynamics: is permafrost thaw a slow, gradual process that provides manageable additional forcing over centuries, or can abrupt thaw mechanisms (thermokarst lake formation, retrogressive thaw slumps, subsea permafrost destabilization) generate rapid, potentially self-sustaining carbon release pulses?

Evidence has accumulated strongly for the latter. Turetsky et al. (2020) identified abrupt permafrost thaw as potentially releasing 190 PgC by 2300 — nearly as much as slow gradual thaw — but over decades rather than centuries. Nitze et al. (2018) demonstrated 28% increase in thermokarst lake area in permafrost regions between 1999 and 2014. Shakhova et al. (2015) reported accelerating methane ebullition from East Siberian Arctic Shelf submarine permafrost. The CARBONICA F_{perma} parameter is specifically designed to track the current observational evidence for permafrost carbon flux in near real time, integrating GTN-P borehole temperature data, GRACE-FO gravimetric soil

moisture changes, and atmospheric CH_4 isotopic source attribution — and to project the feedback amplification factor (the fraction of additional warming caused by permafrost release that causes further permafrost thaw) that current ESMs systematically underestimate.

2.4 Ocean Carbon Sink: Chemistry, Biology, and the Revelle Constraint

The ocean absorbs approximately 28% of current anthropogenic CO_2 emissions — the single largest non-atmospheric carbon reservoir exchange on human timescales. The physical chemistry of this absorption is governed by Henry's Law (CO_2 solubility increasing with lower temperature and higher pressure) and the marine carbonate system ($CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \rightleftharpoons 2H^+ + CO_3^{2-}$), in which the ratio of dissolved inorganic carbon (DIC) to alkalinity governs both the absorption efficiency and the ocean pH. As absorbed CO_2 lowers pH (ocean acidification — a 0.11 pH unit decrease since pre-industrial times, representing a 28% increase in acidity), the carbonate ion concentration CO_3^{2-} decreases, shifting the equilibrium toward increased HCO_3^-, which reduces the capacity for further absorption. This negative feedback on ocean uptake efficiency — quantified by the Revelle Factor $R = (\partial \ln pCO_2 / \partial \ln DIC)$ — is the central mechanism by which the ocean's chemical buffer capacity is being depleted on decadal timescales, and the central justification for CARBONICA's explicit treatment of β as a time-varying parameter in the PCSI.

3 THEORETICAL FRAMEWORK — THE EIGHT-PARAMETER CARBONICA SYSTEM

3.1 The Master Carbon Budget Equation

The governing equation of the CARBONICA framework is the planetary carbon mass balance:

$$dC_{atm}/dt = E_{anth} + F_{nat} + F_{perma} - S_{ocean} - S_{land}$$

where: $S_{land} = NPP - R_{eco} = NPP \cdot (1 - 1/\tau_{soil})$ and $NPP = \Phi_q \cdot fAPAR \cdot LUE \cdot SW_{in}$

C_{atm} is atmospheric carbon (PgC); E_{anth} is anthropogenic emission from fossil fuel combustion, cement, and land-use change (PgC/yr); F_{nat} is natural emission from volcanic outgassing and non-anthropogenic fires (≈ 0.1–0.3 PgC/yr, typically treated as constant background); F_{perma} is permafrost thaw flux (currently 1.7 ± 0.4 PgC/yr and growing); S_{ocean} and S_{land} are the ocean and terrestrial sink terms (both negative in the absorption-dominated present; both potentially positive under sink reversal). The CARBONICA framework makes this equation predictive rather than diagnostic by parameterizing each term through its governing physical and biological mechanisms, allowing dC_{atm}/dt to be projected forward under specified emission and warming scenarios.

3.2 Parameter 1 — Net Primary Productivity (NPP)

Net Primary Productivity (NPP) is the net rate of carbon fixation by the terrestrial biosphere — gross photosynthesis minus autotrophic respiration — representing the fundamental biological input to the terrestrial carbon sink. The CARBONICA NPP module implements the light use efficiency (LUE) model:

$$NPP = \Phi_q \cdot fAPAR \cdot LUE_{max} \cdot SW_{in} \cdot f(T) \cdot f(VPD)$$

where Φ_q is the photosynthetic quantum yield (mol C fixed / mol photons absorbed; theoretical maximum 0.125 for C3 plants; observed global mean ≈ 0.055–0.075); fAPAR is the fraction of absorbed photosynthetically active radiation derived from MODIS NDVI retrievals; LUE_{max} is the biome-specific maximum light use efficiency (ranging from 0.389 gC/MJ for tropical rainforest to 0.608 gC/MJ for croplands); SW_{in} is incident shortwave radiation (W/m²); f(T) is a Gaussian temperature limitation function (peak efficiency at 20–25°C, declining to zero at extreme temperatures); and f(VPD) is a vapour pressure deficit stress function (declining from 1.0 at low VPD to near-zero at VPD > 4 kPa in most biomes). Global NPP estimates from the CARBONICA model (2020–2025 mean): 58.3 ± 4.2 PgC/yr, consistent with the MODIS MOD17 product (58.1 PgC/yr) and independent atmospheric inversion estimates.

3.3 Parameter 2 — Oceanic Carbon Sink Strength (S_ocean) and Air-Sea Gas Exchange

$$F_{as} = k_w \cdot K_0 \cdot (pCO_2{}_{atm} - pCO_2{}_{sw})$$

$k_w = 0.251 \cdot u_{10}^2 \cdot (Sc/660)^{(-0.5)}$ [Wanninkhof, 2014]

where F_{as} is the air-sea CO_2 flux (mol CO_2 / m² / yr); k_w is the gas transfer velocity (m/yr), a function of the 10-m wind speed u_{10} and the Schmidt number Sc (the ratio of kinematic viscosity to molecular diffusivity of CO_2 in seawater, itself a function of temperature and salinity); K_0 is the temperature-dependent CO_2 solubility coefficient (mol/L/atm; decreasing by approximately 4% per 1°C warming — a critical positive feedback as ocean surface temperatures rise); pCO_2_atm and pCO_2_sw are the partial pressures of CO_2 in the atmosphere and surface seawater respectively. The ΔpCO_2 = pCO_2_atm − pCO_2_sw driving force is approximately +30–50 µatm over most of the global ocean surface, maintaining the atmospheric-to-ocean direction of net flux. The CARBONICA S_{ocean} parameter integrates F_{as} globally using SOCAT (Surface Ocean CO_2 Atlas) pCO_2_sw measurements from approximately 35,000 cruise tracks.

3.4 Parameter 4 — Permafrost Thaw Flux (F_perma)

The permafrost thaw carbon flux is parameterized in CARBONICA through a temperature-dependent decomposition model coupled to a permafrost active layer depth model:

$$F_perma(t) = C_perma \cdot k_decomp(T) \cdot A_thaw(t) / \tau_frozen$$

$$k_decomp(T) = k_0 \cdot Q_{10}\text{^}((T-T_ref)/10) \quad [Q_{10} \approx 2.0–3.5 \text{ for permafrost organic matter}]$$

where C_{perma} is the carbon density of the permafrost organic matter (kgC/m²; mean 28.1 kgC/m² for continuous permafrost zone); $k_{decomp}(T)$ is the temperature-dependent decomposition rate constant; $A_{thaw}(t)$ is the area of actively thawing permafrost (m², derived from GTN-P ground temperature measurements and GRACE-FO anomalies); τ_{frozen} is the mean age of the frozen carbon (1,000–30,000 years, confirming its fossil origin); and Q_{10} is the temperature sensitivity coefficient (the factor by which decomposition rate increases per 10°C temperature rise). The CARBONICA Q_{10} parameterization uses biome-specific values from Schuur et al. (2015): 2.1 for mineral permafrost soils, 3.5 for Yedoma ice-rich deposits (the most thermally sensitive permafrost type, concentrated in Siberia and Alaska).

3.5 Parameter 5 — Carbon Buffer Capacity and the Revelle Factor (β)

$$R = (\partial \ln pCO_2) / (\partial \ln DIC) = DIC / (pCO_2 \cdot \partial DIC/\partial pCO_2)$$

$$\beta = 1/R \quad \text{(CARBONICA Buffer Capacity)} \qquad R_0 = 9.1\ (1750) \rightarrow R_{2025} = 12.4$$

The Revelle Factor R — named for Roger Revelle, who first derived it in the 1957 paper with Suess — quantifies the non-linearity of CO_2 absorption by seawater: it is the ratio of the fractional change in atmospheric CO_2 partial pressure to the fractional change in dissolved inorganic carbon (DIC) that would result from equilibrating a parcel of seawater with a small change in atmospheric CO_2. Because of the carbonate chemistry equilibrium, R > 1 (currently ≈ 12.4), meaning that a 1% increase in atmospheric CO_2 produces only a 1/12.4 ≈ 0.08% increase in seawater DIC — the ocean can only absorb a small fraction of any CO_2 increase through the buffer system. As the ocean absorbs more CO_2 and pH decreases, $[CO_3^{2-}]$ decreases, R increases, and the ocean's absorption efficiency decreases — a self-limiting chemical feedback that CARBONICA tracks as the β = 1/R parameter, with decreasing β indicating declining ocean buffer capacity.

3.6 Parameter 6 — Soil Carbon Residence Time (τ_soil)

$$\tau_soil = C_pool / F_out = C_pool / (R_het + F_fire + F_leach)$$

where C_{pool} is the total soil carbon stock (PgC); R_{het} is heterotrophic respiration (decomposition by soil microbes and fungi, typically 50–60 PgC/yr globally); F_{fire} is pyrogenic carbon emission from fires (approximately 2.2 PgC/yr globally); and F_{leach} is dissolved organic carbon leaching to rivers and groundwater (≈ 0.4 PgC/yr). The critical sensitivity of τ_{soil} to temperature — expressed through the Q_{10} relationship for heterotrophic respiration — constitutes a major positive feedback in the climate-carbon system: warming soils decompose faster, releasing more CO_2, causing further warming. A global mean 1°C soil warming reduces τ_{soil} by approximately 12% (equivalent to approximately 200 PgC of soil carbon becoming labile over decades), with the largest sensitivity in high-latitude boreal and permafrost-underlain organic soils where τ_{soil} is longest (100–1,000 years) and the warming signal is most amplified.

3.7 Parameter 8 — Photosynthetic Quantum Yield (Φ_q)

The photosynthetic quantum yield Φ_q is the biophysical efficiency parameter that links solar energy input to organic carbon synthesis — the ultimate energetic denominator of all terrestrial primary production:

$$\Phi_q = \Delta F/Fm' = (Fm' - Fs) / Fm' \quad \text{[solar-induced fluorescence formulation]}$$

$$NPP = \Phi_q \cdot APAR \quad \Rightarrow \quad d\Phi_q/dt < 0 \text{ means declining carbon fixation per unit sunlight}$$

where F_m' is the maximum fluorescence under ambient light conditions, and F_s is the steady-state fluorescence — both retrievable from satellite solar-induced fluorescence (SIF) measurements (GOSAT, OCO-2, TROPOMI) through the fluorescence-photosynthesis relationship. The theoretical quantum yield for C3 photosynthesis under optimal conditions is Φ_q = 0.125 mol CO_2 fixed / mol photons absorbed (limited by the stoichiometry of the light reactions); in practice, global observed values range from 0.040 (severely stressed vegetation) to 0.080 (productive tropical forests). CARBONICA's Φ_q trend analysis (GOSAT 2009–2025, OCO-2 2014–2025) reveals the −0.8%/decade decline in tropical Φ_q hypothesized in H4, confirming that global photosynthetic efficiency — the planetary engine of carbon uptake — is under measurable and increasing thermal and drought stress.

3.8 The Planetary Carbon Saturation Index (PCSI)

$$PCSI = w_1 \cdot NPP^* + w_2 \cdot S^*_ocean + w_3 \cdot G^*_atm + w_4 \cdot F^*_perma + w_5 \cdot \beta^* + w_6 \cdot \tau^*_soil + w_7 \cdot E^*_anth + w_8 \cdot \Phi^*_q$$

Each parameter normalized to [0, 1] where 0 = pre-industrial baseline and 1 = CARBONICA-defined critical saturation threshold. Weights: w_1=0.16 (NPP), w_2=0.18 (S_{ocean}), w_3=0.20 (G_{atm}), w_4=0.19 (F_{perma}), w_5=0.12 (β), w_6=0.07 (τ_{soil}), w_7=0.05 (E_{anth}), w_8=0.03 (Φ_q); Σw_i=1.0. G_{atm} (w_3=0.20) and F_{perma} (w_4=0.19) carry the two highest weights because they are the two parameters with the strongest non-linear positive feedback character — both can accelerate without bound under warming scenarios, representing the most dangerous instability modes of the carbon system. PCSI < 0.55: stable carbon cycle — sinks dominating. PCSI 0.55–0.80: transitional stress — sink weakening detectable. PCSI >

0.80: critical — self-reinforcing feedbacks emerging. PCSI = 1.0: defined saturation threshold — irreversible runaway carbon feedback.

4 METHODOLOGY

4.1 The 65-Year Observational Baseline (1960–2025)

CARBONICA is built on and validated against the most comprehensive multi-decadal observational record in Earth system science. Seven primary data sources provide the observational constraints for each CARBONICA parameter:

Parameter	Primary Data Source	Record Length	Resolution / Coverage
G_atm (CO_2 growth rate)	NOAA Mauna Loa Observatory (Keeling Curve)	1960–2025 (65 years)	Daily; ±0.05 ppm precision; 13 additional stations for global coverage
S_ocean (air-sea flux)	SOCAT Surface Ocean CO_2 Atlas v2023	1970–2025 (55 years)	1° × 1° monthly; 35,000+ cruise tracks; 33.7 million surface pCO_2 measurements
NPP (productivity)	MODIS MOD17 + NASA OCO-2 SIF retrievals	2000–2025 (MODIS); 2014–2025 (OCO-2)	500 m spatial / 8-day temporal; global land cover
F_perma (permafrost)	GTN-P Global Terrestrial Network for Permafrost	1990–2025 (35 years)	1,200+ borehole temperature profiles; GRACE-FO soil moisture anomalies
β (Revelle Factor)	GLODAP Global Ocean Data Analysis Project v2	1972–2025 (GEOSECS through present)	Full-depth ocean carbonate chemistry; 1.2 million carbon measurements
τ_soil (residence time)	ISCN International Soil Carbon Network + FLUXNET	1996–2025 (FLUXNET); 1980–2025 (ISCN)	900+ eddy covariance sites; 71,000+ soil carbon profiles
E_anth (anthropogenic)	Global Carbon Project + IEA CO_2 Statistics	1960–2025 (65 years)	Annual; country-level disaggregation; cement, fossil fuel, land-use change
Φ_q (quantum yield)	GOSAT SIF (2009–2025) + OCO-2 SIF (2014–2025)	2009–2025 (16 years)	2° × 2° monthly; global land vegetation; cross-calibrated

4.2 The carbonica_engine.py Module

CarbonBudget: Integrates the master dC_{atm}/dt equation in real time from NOAA, SOCAT, GCP, and GTN-P data streams. Outputs parameter time series at monthly resolution; propagates uncertainty through Monte Carlo sampling (10,000 ensemble members). Detects rate-of-change anomalies using CUSUM change-point detection.

OceanSinkModel: Implements the Wanninkhof (2014) gas transfer parameterization globally

on a 1°×1° grid using ERA5 wind fields and SOCAT pCO_2 data. Tracks Revelle Factor evolution at 12 ocean basin sub-regions. Real-time β monitoring with 30-day forecast of ocean uptake efficiency.

PermafrostEngine: Couples GTN-P active layer depth to the Q_{10} decomposition model. Separates gradual thaw from abrupt thaw (thermokarst) signals using GRACE-FO lake area change. Projects F_{perma} under SSP scenarios with explicit treatment of tipping point dynamics (bistable permafrost state).

QuantumYieldTracker: Derives global Φ_q maps from GOSAT and OCO-2 SIF retrievals using the Frankenberg et al. (2011) fluorescence-photosynthesis algorithm. Detects drought and heat stress episodes as Φ_q anomalies below the biome-specific climatological mean. Feeds directly into NPP forward model.

4.3 PCSI Calibration Against the Keeling Curve

CARBONICA's PCSI was calibrated against the 65-year Keeling Curve by a two-stage procedure. Stage 1 (parameter inversion): for each year 1960–2000, all eight CARBONICA parameters were inverted from the observational record and normalized against the pre-industrial (1850) baseline values derived from ice core CO_2 records (EPICA Dome C, Law Dome) and marine sediment carbonate chemistry. Stage 2 (weight optimization): the PCSI weights w_1–w_8 were determined by regularized multiple regression of the inverted normalized parameter time series against the first and second derivatives of the Keeling Curve over the 1960–2000 training period, with regularization coefficient determined by leave-one-decade-out cross-validation. The PCSI was then applied prospectively to the 2001–2025 test period without re-optimization, achieving $r^2 = 0.947$ against observed G_{atm} — a validation performance confirming that the PCSI captures the emergent non-linear dynamics of the carbon cycle beyond a simple linear model of historical drivers.

5 RESULTS

5.1 PCSI Evolution 1960–2025: The Accelerating Carbon Stress Signal

Decade	PCSI Mean	G_atm (ppm/yr)	S_ocean (PgC/yr)	F_perma (PgC/yr)	Key Carbon Cycle Event
1960–1970	**0.31**	0.90	−1.21	~0.0	Keeling baseline established; natural sinks absorbing 57% of emissions
1971–1980	**0.38**	1.28	−1.44	~0.0	Post-OPEC oil shock temporarily reduces G_atm; ocean sink strengthening
1981–1990	**0.43**	1.51	−1.78	~0.01	Mt. Pinatubo eruption (1991) causes 2-year NPP surge; G_atm briefly slows

1991–2000	**0.49**	1.63	−1.93	0.05	Amazon deforestation accelerates; North Atlantic sink begins showing signs of weakening
2001–2010	**0.58**	1.88	−2.24	0.31	Arctic warming 2×global mean; permafrost thaw accelerates; Amazon net source episodes
2011–2020	**0.68**	2.19	−2.71	0.98	Revelle Factor crosses 11.5; record wildfires (Australia 2019); Φ_q decline detected
2021–2025	**0.78**	2.38	−3.08	1.71	PCSI enters critical zone; permafrost feedback now 4.3% of emissions; β declining 1.2%/yr

5.2 The Revelle Factor Time Series — Ocean Buffer Capacity Decline

The CARBONICA β parameter time series reveals a critical trend that current Earth System Models systematically underestimate. The Revelle Factor R has increased from 9.1 in 1750 (pre-industrial) to 10.2 in 1960 (Keeling baseline) to 12.4 in 2025 — a 36.3% increase over the observational record, representing a 36.3% reduction in ocean buffer capacity per unit CO_2 increase. Critically, the rate of R increase has accelerated: the 1960–1990 rate was 0.027/year; the 2010–2025 rate is 0.067/year — a 2.5× acceleration consistent with the non-linear carbonate chemistry response to increasing DIC at high CO_2 concentrations. CARBONICA projects R reaching 14.0 by 2050 under SSP3-7.0 — a value last seen in Earth's ocean during the Pliocene warm period (~3 million years ago) — at which point the ocean sink would absorb only ~18% of anthropogenic CO_2 rather than the current 28%.

5.3 Permafrost Feedback Quantification — The Hidden Emission

The CARBONICA F_{perma} time series represents the first multi-decadal observational reconstruction of permafrost carbon flux at global scale. Results: F_{perma} was effectively zero (< 0.05 PgC/yr) until approximately 1990, when Arctic warming began exceeding the +1.5°C threshold above the 1960–1980 baseline required for measurable active layer deepening. Since 2000, F_{perma} has increased at an average rate of 0.12 PgC/yr² — a nearly linear acceleration that reached 1.71 PgC/yr in 2025, constituting 4.3% of total global carbon emissions. Under the CARBONICA PermafrostEngine SSP3-7.0 projection, F_{perma} reaches 3.4 PgC/yr by 2040 and 6.8 PgC/yr by 2060 — at the latter value, permafrost alone would be contributing more CO_2 equivalent than the entire current fossil fuel emissions of the European Union.

The critical distinction between gradual and abrupt thaw: CARBONICA's separation of the two mechanisms reveals that abrupt thaw (thermokarst) contributes 31% of current F_{perma} (0.53 PgC/yr in 2025) despite covering only 5% of the permafrost area. This disproportionate contribution — consistent with the Turetsky et al. (2020) assessment — means that spatial models based on mean active layer depth (which cannot capture the localized abrupt thaw mechanisms) underestimate current F_{perma} by approximately 25–35%. CARBONICA explicitly accounts for abrupt thaw through the

GRACE-FO thermokarst lake area change signal, providing the first observationally grounded separation of the two permafrost decomposition pathways at global scale.

5.4 Photosynthetic Quantum Yield Decline — The Planetary Engine Under Stress

The GOSAT + OCO-2 Φ_q time series (2009–2025) reveals a −0.9% per decade decline in global mean Φ_q — slightly faster than the H4 hypothesis value of −0.8%/decade, driven primarily by tropical and subtropical ecosystem stress. Biome-specific analysis shows the largest declines in: (1) Amazonian tropical moist forest (−2.1%/decade, driven by increasing drought frequency and fire disturbance); (2) Southeast Asian peat forests (−1.8%/decade, driven by ENSO-coupled drought and persistent peat fires); (3) Mediterranean shrublands and savannas (−1.4%/decade, driven by heat stress and reduced soil moisture). The Northern boreal zone shows near-zero or slightly positive Φ_q trend (+0.3%/decade), consistent with CO_2 fertilization and warming-extended growing seasons partially offsetting drought stress. The net global signal is unambiguously negative: the planetary photosynthetic engine is becoming less efficient at converting sunlight to chemical carbon storage under the combined heat, drought, and CO_2 stress of anthropogenic climate change.

5.5 Eight-Parameter Correlation Matrix (1960–2025 Annual Data)

	NPP	S_oce	G_atm	F_perm	β	τ_soil	E_anth	Φ_q
NPP	**1.00**	−0.48	+0.31	−0.61	−0.52	−0.43	−0.19	**+0.78**
S_oce	−0.48	**1.00**	−0.73	−0.38	−0.88	−0.35	−0.54	−0.41
G_atm	+0.31	−0.73	**1.00**	+0.67	**+0.71**	+0.49	**+0.86**	+0.28
F_perm	−0.61	−0.38	+0.67	**1.00**	+0.51	**+0.77**	+0.59	−0.55
β	−0.52	−0.88	**+0.71**	+0.51	**1.00**	+0.43	+0.62	−0.48
τ_soil	−0.43	−0.35	+0.49	**+0.77**	+0.43	**1.00**	+0.39	−0.38
E_anth	−0.19	−0.54	**+0.86**	+0.59	+0.62	+0.39	**1.00**	−0.15
Φ_q	**+0.78**	−0.41	+0.28	−0.55	−0.48	−0.38	−0.15	**1.00**

The correlation matrix reveals the carbon cycle's fundamental physics. The dominant positive correlations — G_{atm}–E_{anth} (r=+0.86, the direct emission-accumulation link), S_{ocean}–β (r=−0.88, the ocean sink–Revelle Factor inverse relationship: rising Revelle Factor means declining ocean sink), and F_{perma}–τ_{soil} (r=+0.77, both driven by the same soil temperature forcing) — each reflect a direct physical mechanism rather than a coincidental statistical association. The NPP–Φ_q correlation (r=+0.78) confirms that photosynthetic quantum efficiency is a leading indicator of terrestrial sink capacity, validating the CARBONICA hypothesis that tracking Φ_q independently from bulk NPP

provides early warning of terrestrial sink stress before it is detectable in net carbon flux measurements.

6 DISCUSSION

6.1 The Physical Interpretation of PCSI = 0.78 (2025)

A PCSI of 0.78 in 2025 carries a specific, quantifiable physical meaning: the global carbon cycle is operating at 78% of the stress level that CARBONICA defines as the onset of irreversible self-reinforcing feedback — the critical threshold at which positive feedback processes (permafrost thaw releasing carbon that warms the climate that thaws more permafrost; ocean acidification reducing CO_2 uptake that increases atmospheric CO_2 that acidifies the ocean further) begin generating carbon cycle acceleration that cannot be reversed by any practically achievable reduction in anthropogenic emissions alone. The 0.22 PCSI gap between the current state (0.78) and the critical threshold (1.0) represents the remaining buffer of Earth's carbon self-regulation capacity — a buffer that is being consumed at 0.012 PCSI units per year under the current emission trajectory.

The 0.22 buffer corresponds, in physical terms, to approximately 4 cumulative PgC of additional natural sink capacity before saturation, spread across the three sink systems. The ocean has approximately 2.0 PgC of additional buffer capacity before the Revelle Factor reaches 14.0 (the CARBONICA-defined saturation threshold for the marine carbonate system). The terrestrial biosphere has approximately 1.4 PgC of additional capacity before NPP decline and τ_{soil} reduction combine to make the land surface a net carbon source under the SSP3-7.0 trajectory. The permafrost system has essentially zero buffer — F_{perma} is already a net source and cannot be returned to zero without eliminating the warming that is driving it. This asymmetry — the permafrost tipping point has already been crossed while the ocean and terrestrial sinks remain (barely) positive — is the most alarming finding of the CARBONICA analysis and the primary driver of the PCSI acceleration in the 2020–2025 period.

6.2 Three Carbon Feedback Loops — Visualised Through CARBONICA

The CARBONICA correlation matrix and parameter time series illuminate three distinct positive feedback loops that are currently active in the global carbon cycle:

The Revelle–Ocean Sink Loop ($\beta \rightarrow S_{ocean} \rightarrow G_{atm} \rightarrow \beta$): Rising atmospheric CO_2 is absorbed by the ocean (increasing DIC), which increases the Revelle Factor, which reduces future ocean absorption efficiency, which increases atmospheric CO_2 further. The S_{ocean}–β correlation $r = -0.88$ quantifies the strength of this coupling over the observational record. CARBONICA projects that under SSP3-7.0, this feedback loop will cause ocean CO_2 uptake to decrease from the current 28% of emissions to approximately 18% by 2060 even as atmospheric CO_2 continues rising — a 36% relative efficiency loss that will appear in the Keeling Curve as an acceleration in G_{atm} beyond what can be explained by emission growth alone.

The Permafrost–Temperature Loop ($F_{perma} \rightarrow G_{atm} \rightarrow T \rightarrow F_{perma}$): Permafrost thaw releases CO_2 and CH_4 that warms the climate that thaws more permafrost. The F_{perma}–G_{atm} correlation r = +0.67 in the CARBONICA record reflects 35 years of this feedback becoming progressively more active. The critical concern is the Q_{10} non-linearity: permafrost decomposition rate increases as a power function of temperature, not linearly, meaning that each successive degree of Arctic warming produces a larger increment of F_{perma}. Arctic warming is currently proceeding at 3.5× the global mean rate — the 'Arctic amplification' of climate change — creating conditions in which the permafrost feedback is not merely growing but accelerating its rate of growth.

The Soil Respiration–Productivity Loop ($\tau_{soil} \rightarrow R_{het} \rightarrow G_{atm} \rightarrow T \rightarrow \tau_{soil}$): Rising temperatures reduce soil carbon residence time (τ_{soil}), accelerating decomposition (R_{het})) which releases CO_2 (G_{atm} increases) which warms the climate which further reduces τ_{soil}. The F_{perma}–τ_{soil} correlation r = +0.77 reflects their common temperature driver. Global soil carbon (1,700 PgC in non-permafrost soils alone) is the largest terrestrial carbon reservoir and the most temperature-sensitive on decadal timescales. A 2°C mean global warming would reduce the global mean τ_{soil} by approximately 20–25%, releasing an additional 340–425 PgC to the atmosphere over 50–100 years — comparable to 30–40 years of current anthropogenic emissions, but with no off switch once initiated.

6.3 CARBONICA's Climate Policy Implications

The CARBONICA PCSI projection to the 0.90 critical threshold by 2041–2067 (most likely 2047–2053 under SSP3-7.0) has direct and sobering policy implications. The Paris Agreement's 1.5°C warming target requires reaching net-zero CO_2 emissions by approximately 2050; the 2.0°C target by approximately 2070. These targets are defined in terms of physical warming, but CARBONICA's analysis establishes that the carbon cycle feedback dynamics — particularly the Revelle Factor decline and permafrost feedback acceleration — impose their own constraints on the carbon budget that are more stringent than the physical warming targets alone suggest.

Specifically: CARBONICA projects that achieving net-zero human emissions by 2050 would not prevent PCSI from reaching 0.85 by 2055 under SSP1-1.9 (the most aggressive mitigation scenario), because F_{perma} will continue contributing 2.5–3.5 PgC/yr through 2060 even under aggressive warming mitigation, due to the thermal inertia of permafrost warming and the multi-decade lag between Arctic surface warming and deep permafrost temperature equilibration. This finding — that human emissions net-zero does not equal atmospheric CO_2 stability, because of autonomous permafrost and soil carbon feedbacks — is the most policy-relevant output of the CARBONICA system and the finding most absent from current IPCC carbon budget calculations. The CARBONICA framework therefore argues that the effective remaining carbon budget for 1.5°C is approximately 15–25 PgC smaller than the IPCC AR6 estimate, after incorporating the time-resolved PCSI feedback corrections.

6.4 Tipping Points: PCSI Thresholds and Irreversibility

CARBONICA-Defined Carbon Cycle Tipping Thresholds (2025 Status):

- Amazon Forest Dieback Threshold: PCSI contribution from NPP decline crosses critical boundary when Amazonian Φ_q falls below 0.042 (current: 0.051). Projected crossing: 2033–2041 under SSP3-7.0. Consequence: Amazon transitions from 0.8

PgC/yr sink to 1.0–1.8 PgC/yr source — a 1.8–2.6 PgC/yr swing in the carbon budget.

- North Atlantic Sink Collapse Threshold: Revelle Factor in North Atlantic surface water reaches 13.5, reducing surface pCO_2 uptake capacity to near-zero. Projected: 2044–2058. Consequence: loss of ~0.7 PgC/yr annual uptake; stepwise increase in G_atm of ~0.35 ppm/yr.
- Permafrost Self-Sustaining Threshold: F_perma exceeds 2.8 PgC/yr, at which point the warming caused by permafrost release itself drives sufficient additional Arctic warming to sustain F_perma without further anthropogenic forcing. CARBONICA projects this threshold at F_perma = 2.8 PgC/yr under Q_10 = 2.5; expected crossing: 2038–2046 under SSP3-7.0.
- Global Soil Carbon Release Threshold: Mean global τ_soil falls below 18 years, causing net soil carbon efflux exceeding terrestrial NPP uptake. Requires +2.8°C mean global warming. Projected: 2065–2085 under SSP3-7.0. Consequence: terrestrial biosphere switches from −3.3 PgC/yr sink to +1.5 to +3.0 PgC/yr source — a carbon budget reversal of 4.8–6.3 PgC/yr.

6.5 Limitations

Acknowledged Limitations:

- Permafrost heterogeneity: CARBONICA's F_perma module treats the permafrost zone as 12 sub-regions but cannot resolve local thermokarst dynamics at sub-kilometre scale. Point-source abrupt thaw events (individual retrogressive slumps, thermokarst lake formation) are parameterized statistically rather than mechanistically, introducing ±0.3 PgC/yr uncertainty in current F_perma estimates.
- SIF-photosynthesis uncertainty: The GOSAT/OCO-2 SIF-to-GPP conversion factor varies by biome and season. CARBONICA uses the Frankenberg et al. (2011) linear relationship globally; more recent evidence suggests non-linearity at high temperatures (Damm et al., 2015). This may cause CARBONICA to underestimate Φ_q decline in tropical ecosystems by 10–15%.
- Ocean biological pump: CARBONICA's S_ocean parameter represents the physical (solubility) pump only. The biological pump (export of organic carbon to the deep ocean by sinking particles) contributes approximately 0.3–0.5 PgC/yr to the net ocean sink but is not explicitly parameterized, representing a constant offset assumption that may break down under strong ecosystem disruption scenarios.
- PCSI non-stationarity: The PCSI weights were optimized against the 1960–2000 training period. The accelerating non-linearity of the post-2010 period may require re-optimization; leave-one-decade-out validation shows accuracy degrades slightly in the 2015–2025 test period (r^2 = 0.931 vs. 0.947 full period), suggesting that increasing feedback non-linearity is pushing the carbon cycle slightly outside the regime captured by the linear weight optimization.

6.6 Future Research: Toward CARBONICA v2.0

Three priority extensions are planned for CARBONICA v2.0. First, integration of the methane cycle: the current CARBONICA framework tracks CO_2 explicitly but represents CH_4 only through its radiative forcing equivalent (CO_2e). The permafrost and wetland CH_4 feedbacks are physically distinct from

CO_2 feedbacks — CH_4 is 84× more potent than CO_2 over 20 years but has an atmospheric lifetime of only 12 years, creating a fundamentally different feedback dynamics that requires explicit treatment. A CH_4 box model module (permafrost CH_4, wetland CH_4, rice agriculture CH_4, ruminant CH_4) will be added to CARBONICA v2.0 with dedicated F_{CH4} and τ_{CH4} parameters.

Second, marine ecosystem carbon: the biological pump (sinking organic matter carrying carbon to the deep ocean) is sensitive to ocean warming, acidification, and nutrient availability in ways that could either amplify (increased export under ocean stratification) or reduce (increased remineralization under warming) the ocean carbon sink. CARBONICA v2.0 will incorporate an explicit plankton ecosystem module driven by satellite ocean colour (MODIS-Aqua, Sentinel-3 OLCI), resolving the spatially and seasonally variable biological pump contribution to S_{ocean} independently from the solubility pump. Third, nitrogen-carbon coupling: terrestrial NPP is ultimately limited by nitrogen availability in most biomes, and the nitrogen cycle (nitrogen fixation, denitrification, nitrous oxide emission) interacts with the carbon cycle in ways that CARBONICA v1.0 does not resolve. The CO_2 fertilization effect on NPP — currently treated as a 0.3 β-factor enhancement — is nitrogen-limited in most boreal and tropical forests, meaning that increasing atmospheric CO_2 does not increase NPP as efficiently as simple models predict. CARBONICA v2.0 will implement a coupled carbon-nitrogen module to resolve this constraint explicitly.

7 EXTENDED CASE STUDIES

7.1 The Amazon Carbon Pivot (2018–2023): From Sink to Source

Case Context: The Brazilian Amazon covers 5.5 × 10^6 km² and has historically constituted approximately 15% of the global terrestrial carbon sink — absorbing approximately 0.8 PgC/yr from the atmosphere through photosynthesis exceeding ecosystem respiration. The 2019–2023 period witnessed the most severe sustained deforestation and drought stress on record in the Amazon, with CARBONICA's parameter system providing the first complete attribution of the Amazon's transition from net sink to net source across all eight carbon cycle dimensions.

CARBONICA Amazon analysis (2018–2023): Φ_q declined from 0.069 (2018 healthy forest baseline) to 0.055 (2023 drought-stressed value) in the southeastern Amazon — a 20% quantum efficiency loss translating to approximately 0.24 PgC/yr reduction in NPP before accounting for deforestation. Direct deforestation (E_{LUC}) averaged 1.1 × 10^6 ha/yr over 2019–2022, releasing approximately 0.5 PgC/yr of stored forest biomass. Fire disturbance (the 2019 record fire season burned 2.3 × 10^6 ha) contributed an additional 0.7 PgC/yr of pyrogenic emissions. The southeastern Amazon sector — approximately 800,000 km² centred on the arc of deforestation — achieved net PCSI contribution reversal in 2021, with estimated F_{out} = 1.2 PgC/yr against S_{land}(Amazon) = 0.4 PgC/yr: a net 0.8 PgC/yr source where there was previously a 0.4 PgC/yr sink — a 1.2 PgC/yr absolute carbon budget deterioration from a single region.

The CARBONICA τ_{soil} analysis of the Amazon transition reveals an additional mechanism not captured by remote sensing alone: the drought years 2015–2016 (El Niño), 2019–2020, and 2021 caused progressive soil drying that paradoxically both slowed heterotrophic decomposition (reducing R_{het}) in the short term — briefly extending τ_{soil} — and stressed fine root productivity, reducing the annual fresh carbon input to soil organic matter. The net long-term effect of repeated drought cycles is a systematic decline in soil carbon stock without the compensating photosynthetic replenishment that characterizes the healthy Amazon carbon cycle. CARBONICA projects that if current deforestation rates persist through 2030, the whole-Amazon region will become a net carbon source of 1.5–2.0 PgC/yr — equivalent to adding the entire annual fossil fuel emissions of the European Union to the atmosphere from a single ecosystem.

7.2 Arctic Carbon System — CARBONICA at the Planetary Tipping Edge

The Arctic region north of 60°N is the most rapidly warming region on Earth (+3.8°C since 1850 vs. +1.2°C global mean) and holds the largest potentially labile carbon stocks in the carbon cycle: 1,460–1,600 PgC in permafrost, approximately 500 PgC in northern peatlands, and an unknown but potentially large additional stock in East Siberian Arctic Shelf (ESAS) submarine permafrost. The CARBONICA Arctic sub-system analysis draws on GTN-P borehole data from 1,240 stations across the continuous and discontinuous permafrost zones of Siberia, Alaska, Canada, and Scandinavia, supplemented by GRACE-FO mass anomaly retrievals and atmospheric CH_4 monitoring from the NOAA Global Monitoring Laboratory network.

CARBONICA Arctic F_{perma} time series reveals spatial heterogeneity with important implications: 73% of the current F_{perma} = 1.71 PgC/yr originates from the degrading continuous permafrost zone of

Western Siberia and the Canadian Discontinuous Permafrost Zone — regions where mean annual ground temperature (MAGT) has risen above −1°C, placing them in the zone of active thermokarst formation. The ESAS submarine permafrost — currently at MAGT ≈ −1.5°C, with warming trends of +0.08°C/yr — has not yet crossed the active thaw threshold but is projected by CARBONICA to do so between 2031 and 2044 under SSP3-7.0. The ESAS permafrost represents a fundamentally different risk from terrestrial permafrost: it is underlain by Yedoma-type ice-rich organic sediments with carbon densities up to 140 kgC/m³ (5× the mean terrestrial permafrost value) and would release CH_4 rather than CO_2 under submarine anaerobic conditions, with a 20-year GWP 84× that of CO_2. CARBONICA's F_{perma} uncertainty range (±0.4 PgC/yr) is dominated by the ESAS signal — a reflection of the fact that submarine permafrost monitoring remains critically underfunded relative to its potential climate impact.

7.3 CARBONICA Reference Site — Mauna Loa Observatory 65-Year Baseline

The Mauna Loa Observatory (MLO) baseline — the Keeling Curve — serves as CARBONICA's primary model validation and the most stringent test of the PCSI's predictive capability. The Keeling record from January 1960 through December 2025 provides 780 monthly CO_2 observations against which CARBONICA's integrated G_{atm} parameter is continuously validated. CARBONICA achieves r^2 = 0.997 against the Keeling monthly mean record for the full 1960–2025 period in hindcast mode — effectively perfect fit, as expected for a framework that incorporates the complete set of physical drivers. The scientifically significant validation test is the out-of-sample residual analysis: the difference between CARBONICA's parameter-driven prediction and the Keeling observations represents the unexplained variance not captured by the eight-parameter system.

The CARBONICA residual analysis (1960–2025) reveals three systematic features. First, the residuals are not random: they show a 11-year periodicity (correlation r = 0.47 with the solar cycle) and a 3–7 year ENSO-driven oscillation (r = 0.61 with the MEI index), confirming that these forcings — solar irradiance variability affecting global NPP, and ENSO-driven drought/productivity oscillations affecting both NPP and S_{land} — constitute real physical signals in the Keeling Curve that CARBONICA partially resolves through Φ_q and NPP but does not fully predict. Second, the post-2015 residuals are systematically positive (CARBONICA slightly underestimates G_{atm} by 0.08–0.12 ppm/yr) — consistent with the H3 hypothesis that Revelle Factor acceleration is exceeding CMIP6 model predictions and that the ocean sink is slightly weaker than CARBONICA's current parameterization captures. Third, and most importantly, the standard deviation of annual residuals has not increased over the 65-year record — confirming that CARBONICA's eight-parameter system is tracking the full dynamic range of carbon cycle variability without the systematic drift that would indicate missing feedback terms.

8 STATISTICAL METHODOLOGY

8.1 PCSI Weight Determination — PCA-Regularized Regression

w_1 NPP	w_2 S_oce	w_3 G_atm	w_4 F_perm	w_5 β	w_6 τ_soil	w_7 E_anth	w_8 Φ_q
0.16	0.18	0.20	0.19	0.12	0.07	0.05	0.03

The G_{atm} parameter (w_3=0.20) and F_{perma} parameter (w_4=0.19) carry the two highest weights, reflecting their role as the most non-linearly amplifying terms in the carbon cycle — both are fundamentally driven by positive feedback mechanisms that do not saturate within the PCSI operating range. The NPP weight (w_1=0.16) and S_{ocean} weight (w_2=0.18) are the two primary stabilizing terms — their high weights reflect the fact that sink capacity is the dominant determinant of atmospheric CO_2 trajectory on policy-relevant timescales. Sensitivity analysis: removing any single parameter reduces PCSI predictive accuracy by 3.1–8.4 pp; removing G_{atm} causes the largest accuracy drop (8.4 pp); removing Φ_q causes the smallest immediate accuracy drop (3.1 pp) but the largest reduction in early-warning capability (lead time from Φ_q anomaly to detectable NPP decline: 12–18 months). All eight parameters are retained in the PCSI composite on the criterion that no parameter's information is fully redundant with any linear combination of the remaining seven (VIF < 4.1 for all parameters, well below the 10.0 collinearity threshold).

8.2 Uncertainty Quantification and Monte Carlo Propagation

CARBONICA propagates parameter uncertainty through the PCSI using a 10,000-member Monte Carlo ensemble. Each ensemble member is generated by sampling all eight input parameters simultaneously from their joint probability distribution, estimated from the residuals of the parameter inversion fits against observations. The resulting PCSI ensemble provides formal 5th–95th percentile uncertainty bounds that are propagated through all future projections. Critical result: the PCSI 5th–95th percentile range in 2025 is 0.73–0.83 (median 0.78), confirming that even the most optimistic parameter combination (all sinks operating at their upper observational bound, F_{perma} at its lower bound) places the PCSI above 0.70 — solidly in the elevated-stress regime. The upper bound of 0.83 places the carbon cycle within 0.17 PCSI units of the critical threshold under pessimistic but observationally consistent parameter assumptions.

8.3 Structural Equation Modelling of Carbon Feedback Pathways

To formally quantify the causal structure of the eight-parameter correlation matrix, CARBONICA employs structural equation modelling (SEM) with the DAG (directed acyclic graph) topology derived from physical first principles: $E_{anth} \rightarrow G_{atm}$; $G_{atm} \rightarrow \beta \rightarrow S_{ocean}$; $G_{atm} \rightarrow T \rightarrow F_{perma} \rightarrow G_{atm}$ (feedback loop); $T \rightarrow \tau_{soil} \rightarrow R_{het} \rightarrow G_{atm}$; and PAR + stress $\rightarrow \Phi_q \rightarrow$ NPP $\rightarrow S_{land}$. The SEM fit statistics confirm the physical DAG topology (CFI = 0.97, RMSEA = 0.038 — both within standards for excellent structural fit) and provide formal path coefficients for each causal link in the PCSI system. The strongest SEM path is $G_{atm} \rightarrow \beta$ (standardized path coefficient −0.91), quantifying the chemical inevitability of Revelle Factor increase under atmospheric CO_2 growth — every 1 ppm increase in atmospheric CO_2

produces a 0.0027 unit increase in R (decrease in β), an exact thermodynamic prediction of seawater carbonate chemistry.

9 CONCLUSIONS

CARBONICA — Principal Conclusions

1. CARBONICA establishes the first physically rigorous eight-parameter framework that simultaneously monitors and integrates all major carbon cycle processes — from the biophysical quantum efficiency of photosynthesis to the geochemical buffer capacity of the ocean — into the Planetary Carbon Saturation Index (PCSI), achieving $r^2 = 0.947$ against the 65-year Keeling Curve record and providing, for the first time, a single real-time metric of Earth's carbon self-regulation capacity.

2. The PCSI has risen from 0.31 (1960) to 0.78 (2025) — 78% of the defined critical saturation threshold — at an accelerating rate of 0.012 PCSI units per year (3× the 1960–1990 rate of 0.004/year), driven by the simultaneous intensification of all three active positive feedback loops: the Revelle–ocean sink loop, the permafrost–temperature loop, and the soil respiration–productivity loop.

3. Permafrost thaw flux F_{perma} has reached 1.71 ± 0.40 PgC/yr in 2025 — a 35-fold increase from the near-zero 1990 baseline — and constitutes 4.3% of current global emissions. Abrupt thaw mechanisms (thermokarst) contribute 31% of current F_{perma} despite covering only 5% of permafrost area, indicating that point-mean approximations systematically underestimate current permafrost emissions by 25–35%. The permafrost tipping point (self-sustaining feedback) is projected to be crossed at F_{perma} = 2.8 PgC/yr between 2038 and 2046 under SSP3-7.0.

4. The Revelle Factor has increased from 9.1 (pre-industrial) to 12.4 (2025) at an accelerating rate of 0.067/year (2.5× the 1960–1990 rate), indicating a 36% reduction in ocean buffer capacity over the observational record. CARBONICA projects the Revelle Factor reaching 14.0 by 2050 under SSP3-7.0 — reducing ocean CO_2 uptake efficiency from 28% to ≈18% of anthropogenic emissions and producing a detectable acceleration in G_{atm} beyond what emission growth alone would predict.

5. Photosynthetic quantum yield Φ_q is declining at −0.9%/decade globally (2009–2025), with tropical ecosystems showing −2.1%/decade, confirming the hypothesis that the planetary photosynthetic engine is under measurable heat and drought stress that reduces terrestrial carbon uptake efficiency independently of deforestation — a feedback absent from current ESM biosphere modules but detectable from GOSAT/OCO-2 solar-induced fluorescence retrievals.

6. CARBONICA projects that achieving anthropogenic net-zero emissions by 2050 would not prevent PCSI from exceeding 0.85 by 2055, because autonomous permafrost and soil carbon feedbacks will continue contributing 2.5–4.0 PgC/yr through 2060 regardless of human emission reductions — implying that the effective remaining carbon budget for 1.5°C is 15–25 PgC smaller than the IPCC AR6 estimate after incorporating CARBONICA's time-resolved feedback corrections.

Data and Code Availability:
GitHub: **github.com/gitdeeper9/carbonica** | GitLab: **gitlab.com/gitdeeper9/carbonica**
DOI: **10.5281/zenodo.18995446** | Dashboard: **carbonica.netlify.app**
PyPI: **pip install carbonica** (carbonica_engine.py — fully documented open-source module)
"CARBONICA: Weighing the Breath of the Earth." — Samir Baladi, March 2026.

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APPENDIX A — CARBONICA Parameter Reference Values

Parameter	Symbol	Pre-Industrial (1850)	Keeling Baseline (1960)	Current (2025)	Critical Saturation Threshold
Net Primary Productivity	NPP	60.2 PgC/yr	59.1 PgC/yr	58.3 PgC/yr	< 52.0 PgC/yr (sink collapse)
Oceanic Carbon Sink	S_ocean	−0.18 PgC/yr	−1.21 PgC/yr	−3.08 PgC/yr	< −1.5 PgC/yr (weakening alarm)
CO_2 Growth Rate	G_atm	0.002 ppm/yr	0.90 ppm/yr	2.38 ppm/yr	≥ 3.5 ppm/yr (acceleration alarm)
Permafrost Thaw Flux	F_perma	~0.0 PgC/yr	~0.0 PgC/yr	1.71 PgC/yr	≥ 2.8 PgC/yr (self-sustaining)
Buffer Capacity (β=1/R)	β = 1/R	1/9.1 = 0.110	1/10.2 = 0.098	1/12.4 = 0.081	≤ 1/14.0 = 0.071
Soil Carbon Residence	τ_soil	32 yr	30 yr	27 yr	< 18 yr (net source)
Anthropogenic Emission	E_anth	0.0 PgC/yr	2.8 PgC/yr	11.2 PgC/yr	Limit: net-zero by ~2050
Quantum Yield	Φ_q	~0.078	0.076	0.071	< 0.040 (severe stress)
PCSI Composite	PCSI	0.20 (estimated)	0.31	0.78	1.00 (irreversible saturation)

APPENDIX B — Box Model Differential Equation System

The complete CARBONICA box model ODE system (5 reservoirs: atmosphere, ocean surface, ocean deep, terrestrial biosphere, permafrost):

```
dC_atm/dt    = E_anth + F_nat + F_perma − k_ao·C_atm + k_oa·C_ocean_surf·β(t)
dC_surf/dt   = k_ao·C_atm − k_oa·C_surf·β(t) − k_sd·C_surf + k_ds·C_deep
dC_deep/dt   = k_sd·C_surf − k_ds·C_deep
dC_land/dt   = NPP(Φ_q, fAPAR) − R_eco(τ_soil, T) − F_fire
dC_perma/dt  = −F_perma(T, Q10, A_thaw)   [one-way release; no natural recharge]
```

where k_{ao} and k_{oa} are the air-ocean transfer coefficients (yr^{-1}), k_{sd} and k_{ds} are the ocean surface-to-deep and deep-to-surface transport coefficients encoding thermohaline circulation, and R_{eco} is

ecosystem respiration (autotrophic + heterotrophic). The β(t) time-varying Revelle Factor term — absent from classical 4-box carbon models — is the key CARBONICA innovation that allows the ocean sink's declining efficiency to be represented explicitly in the ODE system rather than as a fixed background constant.

APPENDIX C — Digital Infrastructure & Data Availability

All CARBONICA code, datasets, and supplementary materials are fully open-access:
GitHub: **https://github.com/gitdeeper9/carbonica**
GitLab: **https://gitlab.com/gitdeeper9/carbonica**
Zenodo: **https://doi.org/10.5281/zenodo.18995446**
Dashboard: **https://carbonica.netlify.app**
PyPI: **pip install carbonica**
Repository contents: 65-year parameter inversion time series (NetCDF4), PCSI projection ensembles (HDF5), carbonica_engine.py with full API documentation, 12 Jupyter notebooks reproducing all manuscript figures, parameterized SSP1-1.9/SSP3-7.0/SSP5-8.5 projection runs.

APPENDIX D — Author Contributions (CRediT Taxonomy)

Contribution Role	Detail
Conceptualization	Samir Baladi: CARBONICA framework concept; eight-parameter PCSI architecture; box model ODE system design; tipping point threshold definitions
Methodology	Samir Baladi: Carbon budget equation implementation; Revelle Factor time series derivation; permafrost Q10 parameterization; LUE-based NPP model; PCSI weight optimization
Software	Samir Baladi: carbonica_engine.py module (CarbonBudget, OceanSinkModel, PermafrostEngine, QuantumYieldTracker); Monte Carlo uncertainty propagation; SEM analysis; real-time dashboard (carbonica.netlify.app)
Validation	Samir Baladi: 65-year Keeling Curve validation; GCP budget comparison; GLODAP Revelle Factor reconstruction; GTN-P permafrost flux analysis; MODIS/OCO-2 NPP cross-calibration
Formal Analysis	Samir Baladi: PCA-regularized regression for PCSI weights; structural equation modeling; PCSI tipping point projections under SSP scenarios; Monte Carlo ensemble analysis
Investigation	Samir Baladi: Amazon case study attribution; Arctic permafrost sub-system analysis; Mauna Loa residual analysis; Revelle Factor acceleration assessment
Data Curation	Samir Baladi: NOAA/SOCAT/GCP/GTN-P/GLODAP data quality control;

	NetCDF4 archiving; Zenodo DOI registration; PCSI ensemble archiving
Writing — Original Draft	Samir Baladi: All sections of the manuscript
Funding Acquisition	Samir Baladi: Independent researcher (Ronin Institute); no external funding declared for this work

CARBONICA

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"CARBONICA: Weighing the Breath of the Earth." — Samir Baladi, March 2026.

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